POJITHA HETTIARACHCHILALAGE

IoT, Embedded & AI Engineer

Coventry, UK | +44 7778 053664 | pojithah99@gmail.com

linkedin.com/in/pojitha-hettiarachchilalage-830023161 | pojithatech.co.uk | github.com/pojitha99

PROFESSIONAL SUMMARY

IoT, embedded and AI engineer with 3+ years taking connected products from concept to deployment, from PCB and firmware to embedded Linux, cloud and on-device machine learning. Delivered 4 production-ready IoT/OT products, including devices tested and deployed in trains and train control centre networks. Writes C/C++, Python and Rust for Arm Cortex-M, ESP32 and Raspberry Pi platforms, with FreeRTOS and Yocto-based embedded Linux. Integrates devices with AWS and Azure IoT, and deploys ML for anomaly detection and predictive maintenance at the edge. Strong security background, with secure boot, hardening and firmware analysis experience. MSc Artificial Intelligence (Distinction), BSc (Hons) Computer Security, and 3 research publications.

CORE SKILLS

- **Languages:** C, C++ (object-oriented), Python, Rust, Bash, JavaScript (React, Node.js)
- **Embedded & Firmware:** Arm Cortex-M, ESP32, PIC, Raspberry Pi CM4 / Pi 4 / Pi 5 / Pi Zero 2W, NVIDIA Jetson, FreeRTOS, Bare-metal, Drivers
- **Embedded Linux:** Yocto / OpenEmbedded, BitBake, Buildroot, U-Boot, BSP Integration, Secure Boot, Boot-time Optimisation, systemd
- **Protocols:** SPI, I2C, UART, CAN, Modbus, MQTT, BLE, Ethernet, TCP/IP, DHCP, TFTP, SNMP, Cellular (LTE/5G, PPP), Zigbee
- **IoT & Cloud:** AWS IoT Core and Greengrass, Amazon S3, Azure IoT Hub and IoT Central, Event Grid, Terraform, Docker
- **AI & ML:** Machine Learning, Deep Learning, TensorFlow, On-device ML Inference, Anomaly Detection, Predictive Maintenance, Sensor Fusion
- **Hardware & Debug:** PCB Design (KiCad, Fusion 360), Power Management, Signal Isolation, Oscilloscope, Logic Analyser, JTAG/SWD, Wireshark
- **Security & Practices:** System Hardening, HashiCorp Vault, Vulnerability Assessment, Firmware Analysis (Ghidra), IEC 61508 / ISO 26262, Git, GitLab CI/CD

PROFESSIONAL EXPERIENCE

IoT Hardware Engineer, Complete Cyber Ltd, Harwell Science Campus | Nov 2023 - Sep 2026

- Delivered 4 production-ready IoT/OT products from concept to deployment, including an edge security gateway and power monitoring solutions tested and deployed in trains and train control centre networks.
- Designed a custom PCB for a two-board Raspberry Pi device in KiCad and Fusion 360, integrating battery management, MOSFET power switching, optocoupler isolation, GNSS and a SIM7600 cellular modem, and brought it up with oscilloscope and logic analyser.
- Developed EcoBeacon, a power monitoring device measuring AC/DC voltage, current and temperature, written in object-oriented C++ on ESP32 and Raspberry Pi Zero 2W with FreeRTOS and LTE/5G connectivity.
- Engineered Yocto-based embedded Linux builds for Raspberry Pi CM4 and Arm systems with secure boot, U-Boot configuration, BSP integration, boot-time optimisation and system hardening.
- Developed C and Python drivers for Arm Cortex-M and ESP32 over SPI, I2C, UART, CAN, Modbus, MQTT, BLE and PPP.
- Deployed on-device ML pipelines for anomaly detection and predictive maintenance, and moved the product to a faster platform when the original hardware couldn't meet real-time targets.
- Built device-to-cloud pipelines that encrypted data (JWE) and uploaded it to AWS, with Greengrass device authentication, fleet monitoring and remote debugging, provisioned via Terraform.
- Designed a one-way hardware data diode using optocoupler-isolated UART and a CRC16-CCITT framed protocol.
- Automated OT/IT security assessments by generating 223,000+ custom vulnerability plugins from CVE datasets.
- Presented project concepts at IDEALIST 2024 and represented the company at TechEx and Rail Forum events.

Embedded Software Engineer, Faraday Battery, Birmingham | Sep 2023 - Nov 2023

- Implemented BLE security for battery management systems, including PassKey authorisation, secure pairing and encrypted key exchange.
- Tested code and analysed vulnerabilities using Polyspace static analysis in line with IEC 61508 / ISO 26262.
- Managed BLE data transmission with persistent SQLite logging for reliable data integrity.

Artificial Intelligence Lecturer, Genius Bit Academy (Online) | Apr 2025

- Designed and delivered applied AI/ML training to 100+ students, achieving a 100% pass rate on final projects.

Intern IoT Developer, Arimac Digital | Oct 2021 - Jul 2022

- Developed and deployed Azure IoT infrastructure (IoT Central, IoT Hub, telemetry, Event Grid) for monitoring and controlling humanoid robots.
- Developed a stock management system and API web app with React and Node.js, and managed GitLab CI/CD pipelines.

PROJECTS

- **iTED Glass:** wearable thermal imaging spectacles on NVIDIA Jetson with a Seek Thermal camera for real-time contactless temperature detection (2020 Hackadev COVID-19 Challenge).
- **AI Agent for Cybersecurity (in progress):** AI/ML-driven agent for bug bounty vulnerability discovery.
- **Elderly Care Monitoring Device:** designing a Raspberry Pi 4-based device, with a Home Assistant setup sending fall and location alerts to family.
- **Custom CM4 Carrier Board:** redesigning a dual-Ethernet Raspberry Pi CM4 carrier board in Fusion 360 Electronics.

PUBLICATIONS

- Identify and Helps to Manage Human Mental Fatigue Using AI, MSc Thesis, 2023. DOI: 10.13140/RG.2.2.22829.87521
- Machine Learning Techniques for Intrusion Detection and Bandwidth in Network System using CIC-IDS-2018, Technical Report, 2023. DOI: 10.13140/RG.2.2.29099.52000
- Secure IoT Infrastructures through Honeypot, Technical Report, 2022. DOI: 10.13140/RG.2.2.22335.18084

EDUCATION

MSc Artificial Intelligence (Distinction), Birmingham City University | 2022 - 2023

BSc (Hons) Computer Security, 2:1, University of Plymouth | 2018 - 2021

CERTIFICATIONS

AWS Solutions Architect | Amazon S3 | Cisco IoT Fundamentals | Cisco Introduction to IoT | Machine Learning using Python | Python for Data Science | ISC2 Cybersecurity (in progress)